Chapter 9, Problem 92CP

Problem

A transmission line has a series impedance of $\mathbf{Z} = 100\,\underline{/75^\circ}\ \Omega$ and a shunt admittance of **Y** = $450\,\underline{/48^\circ}\ \mu S.$ Find: (a) the characteristic impedance $\mathbf{Z}_o = \sqrt{\mathbf{Z}/\mathbf{Y}}$, (b) the propagation constant $\gamma = \sqrt{\mathbf{ZY}}$.

Step-by-step solution

Step 1 of 2

(A) $Z_0 \quad = \quad \sqrt{\dfrac{Z}{Y}}$

$= \quad \sqrt{\dfrac{100\angle 75^\circ}{450\angle 48^\circ \times 10^{-6}}}$

$= \quad 471.4\angle 135^\circ\ \Omega\quad .$

Step 2 of 2

(B) $Y \quad = \quad \sqrt{ZY} \quad = \quad \sqrt{\dfrac{100\angle 75^\circ}{(450\angle 48^\circ \times 10^{-6})^{-1}}}$

Or, $= \quad \sqrt{100\angle 75^\circ \times 450\angle 48^\circ \times 10^{-6}}$

$= \quad 0.2121\angle 61.5^\circ\quad .$